INDUSTRIAL SAFETY INTELLIGENCE

THE THIRD EYE

TRINETRA

The danger no single sensor can see.

On **13 January 2025, eight workers died** in a coke-oven-battery explosion at the Visakhapatnam Steel Plant. The gas sensors had data. The permits were logged. SCADA was running. **No layer connected those signals in time.** Trinetra is the missing intelligence layer that fuses gas, permits, CCTV and shift logs into one real-time brain — and catches the lethal combinations every single system rates "normal," **minutes before they kill.**

100%
COMPOUND-HAZARD RECALL

0%
FALSE-POSITIVE RATE

+7.4 min
MEAN EARLY-WARNING

Saud Satopay · solo build
github.com/SaudSatopay/trinetra

A deterministic, auditable compound-risk engine for zero-harm operations.

Data present. Unacted upon.

Fatalities in heavy industry are rarely one sensor screaming. They are a **combination** that each individual system rates as normal — rising gas still below its alarm, a routine hot-work permit, a crew in a confined space. Every silo is green. The plant is already lethal. The bottleneck is not missing sensors; it is the missing layer that reads them **together.**

8	~3 / day	0
WORKERS KILLED · VIZAG COKE-OVEN BATTERY · JAN 2025	DEATHS IN INDIA'S REGISTERED FACTORIES (DGFASLI)	LAYERS CONNECTED THE SIGNALS IN TIME

> *In more than 30 years in gas and heavy industry, the worst incidents were never one single failure. They were three or four small things lining up at the same time, each looking acceptable on its own. Most of the time, nothing actually flagged that combination before it became a serious event.*
>
> NISHAT MULLA · PLANT MANAGER · 30+ YEARS ACROSS GAS & HEAVY INDUSTRY

ANATOMY OF THE SILENT WINDOW — VIZAG, RECONSTRUCTED

t + 0
Permit opens
Hot-work permit issued in the coke-oven battery. An ignition source, by the book.

→

t + 3
Gas rises
Methane climbs — but stays below every single-sensor setpoint. All clear.

→

t + 5
Crew enters
Three workers in the confined space beneath. Each system, in isolation, normal.

→

t + 14
First alarm
A single gas sensor finally trips — minutes too late to clear the floor.

Three green lights. One lethal combination.

Trinetra's core insight is that danger lives in the **interaction** of signals, not in any one of them. Three readings, each individually acceptable, multiply into a hazard no threshold dashboard will ever flag:

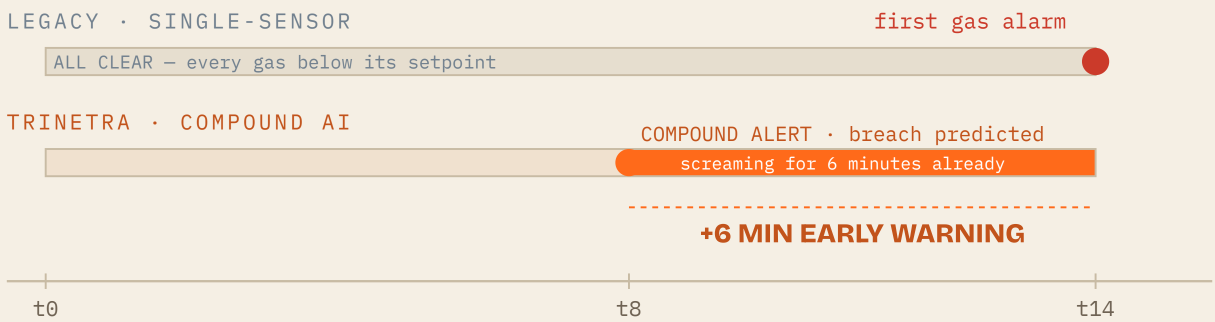
Rising flammable gas
Climbing, but still below its alarm setpoint.
READS NORMAL

Active hot-work permit
A live ignition source in the zone.
READS NORMAL

Personnel present
A crew inside the confined space.
READS NORMAL

= **A lethal combination no single sensor flags** — detected, and acted on, minutes before it becomes critical.

THE MOMENT THAT MATTERS — LEGACY VS. TRINETRA ON THE VIZAG RECONSTRUCTION



WHY IT MATTERS

Six minutes is the window an operator gets to suspend the permit and clear the floor before ignition — while every gas sensor still reads green. The decision is made by a transparent, deterministic engine; the language model never makes the life-safety call.

A deterministic brain that fuses the signals.

A hybrid by design. A reproducible, baseline-subtracted scoring engine makes every safety decision; the language model only explains it, retrieves precedent, and drafts the report. Each frame flows through one pipeline:

01		02		03		04		05
Sense	→	**Fuse**	→	**Predict**	→	**Prescribe**	→	**Act**
gas · permits CCTV · shift logs		compound-risk scoring		time-to-threshold extrapolation		ranked counter-factual action		alert · report evacuate

Compound-risk fusion

Catches danger **below** single-sensor thresholds — context multiplies the gas signal.

Predictive time-to-threshold

Extrapolates the trend to a live breach countdown. Anticipatory, not reactive.

Prescriptive intervention

Ranks the single most effective action by counterfactual risk reduction.

Cross-zone blast radius

Ignition and personnel propagate to adjacent zones via the zone-adjacency graph.

Disaster-memory RAG

Matches live conditions to a corpus of 8 real disasters — an 81% Vizag match.

Autonomous response

Drafts an OISD/Factory-Act report and a multilingual evacuation alert.

WHY HYBRID

Industry juries rightly distrust a language model making a life-safety call. So the decision is a transparent, reproducible score with explicit thresholds — auditable by a safety officer, defensible in a statutory review. If the model is down, the safety call is unaffected.

A connector, not a rewrite.

The digital twin and a real plant feed are the **same interface** — the engine cannot tell the difference. Real-plant data enters the deterministic core through a connector: a SCADA/permit CSV, or a live OPC-UA / MQTT historian feed.

DATA SOURCES · THE PLANT FLOOR

Gas sensors
CH_4 · CO · H_2S · O_2
4–20 mA / OPC-UA

Permit-to-work
hot-work · confined
HSE system

CCTV
person · zone-intrusion
RTSP → YOLOv8

SCADA historian
tags · trends
OPC-UA · MQTT · PI

Shift logs
crew · handover roster

↓

INGEST · DIGITAL TWIN / CONNECTOR

Edge pre-filter → normalise → one unified **PlantSnapshot** per minute. The twin and a live feed are the same interface — **/api/ingest** (CSV) · **/api/opcua/session** (live wire).

↓

THE COMPOUND-RISK ENGINE · THE CORE IP

Deterministic compound scoring

01 **Baseline subtract** → 02 **Feature extract** → 03 **Compound score** → 04 **Predict + prescribe** → 05 **Verdict gate**

score = 100 · raw · (1 + **ignition** + **personnel** + **confined**) → verdict · lead-time · ranked intervention · blast-radius

■ DETERMINISTIC · AUDITABLE · REPRODUCIBLE AT A FIXED SEED · NO LLM IN THE DECISION

LLM-ASSIST SIDECAR

enriches the verdict, never makes it

- Vision — YOLOv8 person / zone-intrusion
- Disaster-Memory RAG — Gemini · 81% Vizag
- 6-stage reasoning graph — LangGraph
- Compliance + shift-left permit gate
- Pre-mortem hazard discovery
- Knowledge graph — networkx · 27 / 36

↓

RESPONSE ORCHESTRATOR

Incident report
Factory Act §36/37/38 · OISD-STD-105

Multilingual alert
English · Telugu · Hindi

Multi-channel dispatch
PA · SMS · mobile · SCADA

Evidence freeze
sensor + CCTV snapshot

↓

CONTROL ROOM

Next.js HMI — split-reality · plant · fleet · knowledge graph · served over **FastAPI** — 30 REST endpoints + a WebSocket stream.

Stack Python · FastAPI · LangGraph · YOLOv8 · Gemini · networkx · Next.js · TypeScript

O(zones) / frame · ~50 µs p50 · 100 / 0 / +7.4 · seed-reproducible

The whole picture.

The same system at a glance — three layers: a digital-twin (or live-connector) **data** layer, a deterministic-first **intelligence** layer, and a real-time control-room **presentation** layer. The decision is a transparent score; the language model only explains it.

INTELLIGENCE LAYER · LLM-ASSISTED · EXPLAINS, NEVER DECIDES

ENRICHES THE VERDICT — IT DOES NOT MAKE IT

Vision (YOLOv8) · Disaster-Memory RAG (Gemini) · 6-stage reasoning graph (LangGraph) · Compliance & permit-gate · Pre-mortem · Knowledge graph (27 nodes / 36 edges)

Data sources

Gas sensors
Permit-to-work
CCTV · shift logs
SCADA historian

→

Ingest

OPC-UA · MQTT
CSV connector
Edge pre-filter
→ digital twin

→

Compound engine

gas level + trend + ignition + personnel + blast-radius · O_2 gate
→ verdict · lead · action

→

Response

Incident report
Multilingual alert
Multi-channel dispatch
Evidence freeze

→

Control room

Next.js HMI split-reality view
plant · fleet
FastAPI REST + WS

DETERMINISTIC · AUDITABLE · NO LLM IN THE DECISION

30 REST endpoints + a WebSocket stream · O(zones) per frame · reproducible at a fixed seed

THE INTEGRATION STORY

Trinetra sits over the sensors a plant already has. It ingests standard SCADA / IoT / permit formats, so going from digital twin to a live plant swaps the data source — not the brain. Proven by replaying two real public incident inquiries and a real third-party measured dataset through the same untuned engine.

Measured, not claimed.

100%
COMPOUND RECALL · 14/14

0%
FALSE-POSITIVE · 0/15

+7.4
MIN MEAN LEAD

Across **29 labelled scenarios** – 14 genuine compound hazards and 15 hard negatives – Trinetra caught every real hazard with zero false alarms, a median **6** and up to **12** minutes ahead of the legacy system. The hardest negatives carry all three explosion factors yet are physically safe (inerted, no oxidizer); held as true negatives on physics, not rules. Deterministic at a fixed seed – every figure here reproduces.

ABLATION — CONTEXT IS WHAT EARNS THE PRECISION (FALSE-ALARM RATE AT THE SAME 7.4-MIN LEAD)

Single-sensor threshold the incumbent	73%
Gas-trend rule early, but context-blind	67%
Trinetra full compound fusion	0%

HELD-OUT GENERALIZATION

100% recall · **3.3%** FP

240 randomized scenarios at unseen simulator seeds the thresholds were never tuned on. Answers the “train = test” critique – it is not overfit to the curated 29. Plus 10/10 fault-mode checks and 5/5 property-based invariants.

REAL INCIDENTS & REAL DATA — SAME ENGINE, NO TUNING

+10 Texas City · **+36** Jaipur

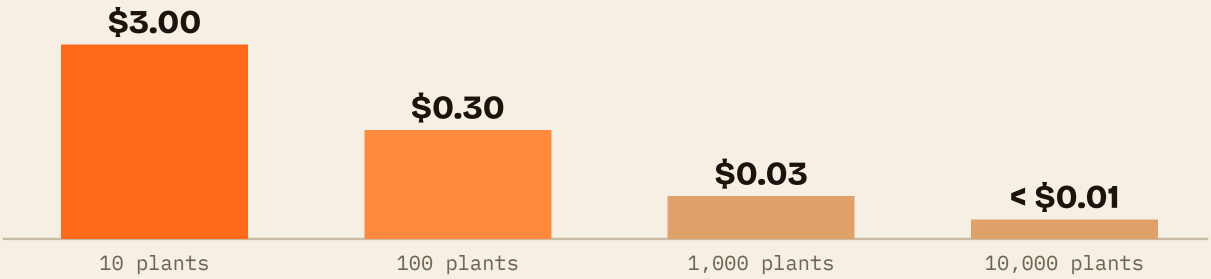
Reconstructed from the U.S. CSB (2005) and MB Lal (2009) inquiries – fires minutes before each documented ignition. And on **real measured CO** (UCI #360, De Vito 2008) and a modeled **EPA ALOHA** methane release the engine never authored, the compound alert holds while a single 10%-LEL sensor stays blind.

One engine. Every plant. Measured.

Real-time on commodity hardware, and the unit economics fall with the fleet. Both are measured live, not asserted — the engine is **O(zones)** and stateless per plant, so the fleet scales horizontally by adding plain workers.

~50 µs	**~0.7M**	**~70×**	**$0.30**
P50 FRAME → ALERT · SINGLE CORE, NO GPU	SENSOR-TAGS / SEC ON ONE CORE	REAL-TIME FOR A 10,000-TAG PLANT	PER PLANT / MONTH AT 100 PLANTS

MEASURED $/PLANT/MONTH — FALLS WITH SCALE (ONE ENGINE, ~5,000 PLANTS PER CORE)



DEPLOYS OVER

- Steel — coke ovens · blast furnaces
- Refining & petrochem — OISD permit regimes
- Mining — confined-space & gas
- Power & chemicals

STATELESS SHARDS

No shared state, no per-site model. A live OPC-UA bridge and the CSV connector already prove that ingesting real plant data is a connector, not a rewrite. The fleet board triages every site compound-first on one pane of glass — no per-site retraining.

The math a buyer underwrites.

PER PREVENTED INCIDENT

₹115.5 Cr

avoided loss — a Vizag-class event.

Lives protected	₹7.5 Cr
Asset damage	₹40 Cr
Downtime (21 d)	₹63 Cr
Regulatory penalty	₹5 Cr

BUYER
Plant process-safety head — owns permit-to-work and statutory compliance, **carries the fatality risk.**

WEDGE
Permit-to-work intelligence — one fundable pain — then expand to full gas + permit + CCTV fusion.

PRICING
~₹1 Cr / plant / yr — versus a single downtime-day (~₹3 Cr) or one OISD / NGT penalty. **~7.7×** expected annual return even at a conservative 1-in-15-year event, plus a recurring insurance-premium offset.

TRIGGER
Post-Vizag / LG-Polymers regulatory pressure — DGMS scrutiny and mandatory near-miss reporting.

Incumbents sell silos

Honeywell, Dräger and MSA sell gas, permits and CCTV separately. The danger lives in the seam no one owns end to end.

Deterministic & audit-grade

Compound rules to real OISD / Factory-Act thresholds — domain IP a safety officer can defend in an audit, not a clonable model.

A per-plant data moat

Operator feedback tunes each site's nuisance profile — proprietary data that compounds per site and raises switching cost.

THE ASK

1–2 design partners (steel / refining) for a 90-day pilot on a live permit-to-work + gas feed.

Three workers a day. The data to stop it already exists. Six minutes is the difference between an evacuation and a funeral — and Trinetra is the layer that acts on it. Before, not after.

Sources: CSB Report 2005-04-I-TX (Texas City); MB Lal Committee (Jaipur, 2009); De Vito et al. 2008, UCI #360 (DOI 10.24432/C5K603); EPA/NOAA ALOHA 5.4.7; DGFASLI / Ministry of Labour. Field validation: an industry veteran, in his own words — problem validation, not a customer endorsement. Every metric reproducible from the runnable harnesses in the repository.

• TRINETRA · SAUD SATOPAY